PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Effectiveness of robotic-assisted therapy for upper extremity function in children and adolescents with cerebral palsy: a systematic review protocol
AUTHORS	Sung-U, Sasithorn; Nisa, Badur; Yotsumoto, Kayano; Tanemura, Rumi

VERSION 1 – REVIEW

REVIEWER	Chris Jones Brighton and Sussex Medical School
REVIEW RETURNED	16-Oct-2020

GENERAL COMMENTS	Study title: "Effectiveness of robotic-assisted therapy for upper extremity in children and adolescents with cerebral palsy: a systematic review protocol" - It seems to me there is a word missing between "extremity" and "in children" - upper extremity what in children? Function? Same for wherever else this phrase appears, such as the abstract. Outcomes and statistics: What kinds of outcomes are expected? It's stated that "outcomes will be divided into two domains of the ICF-CY framework: (1) body function and structure and (2) activity and participation." but what are these likely to be measurements of specifically? How will different types of outcomes be analysed together? More detail needs to be given regarding the statistical analyses - particularly in relation to the stated limitation of the study "We anticipate that meta-analysis could be challenging due to the methodological heterogeneity of feasibility studies." Generally the standard of written English is acceptable, but there are a few instances where sentences contain errors or have unclear meanings. A non-exhaustive list of examples: "Nonetheless, while the independence in activities and social participation of children and adolescents with CP is inevitably affected by motor dysfunction,10 11 eighty percent of those having motor impairments of upper extremity" (end of the sentence doesn't follow from the start) "Summarize of these reviews laid understanding..." (Summaries) "Nevertheless, the extent of service in the perspective of clinical intervention is still unclear." (service seems like the wrong word to use here, so the meaning of the sentence is unclear).

REVIEWER	James Slaughter Vanderbilt University, United States
REVIEW RETURNED	21-Oct-2020

GENERAL COMMENTS	The article clearly describes a protocol to be followed for a systematic review of robtic-assisted therapies for individuals with CP. If the authors have done an initial search, it would be useful to know approximately how many potential studies have been identified for inclusion from the searches. While this initial number will likely be decreased after thorough examination of each study, it would be useful to know the scope of the initial search is neither too large or too small.

REVIEWER	Hans Forssberg Karolinska Institutet Stockholm Sweden
REVIEW RETURNED	31-Oct-2020

GENERAL COMMENTS	The authors are planning a systematic review on robotic assisted therapy for upper extremities in children/youth with all types, and all severity levels of cerebral palsy. The study protocol is well designed and follow the book with all required information, and the manuscript is well written. I cannot find any flaw. My only concern is the introduction which lacks clinical context. Unfortunately, this is quite common in medical engineering. The introduction describes the CP condition from a rather “high level” with proper references, but gives little information about the clinical symptoms and functional limitations that the robot-assisted intervention is supposed to address. There is neither any information on other therapies used to improve motor function in the upper extremities, although robots are often used in combinations with other modes of intervention. In particular, it would be important to mention that much of the evidence based interventions are based on motor learning and intensive, repetitive self-initiated movement, and in which way the use of robots can contribute to improvement.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1: Dr. Chris Jones
Comment 1: Study title: "Effectiveness of robotic-assisted therapy for upper extremity in children and adolescents with cerebral palsy: a systematic review protocol"
- It seems to me there is a word missing between "extremity" and "in children" - upper extremity what in children? Function?
- Same for wherever else this phrase appears, such as the abstract.

Response: Thank you very much for comment and pointing this out. We agree that a word “function” should be added after “upper extremity” to embed a meaningful term.
We have carefully checked and specified it in the title (p. 1), abstract (p. 2), objectives (p. 7), and content (p. 6, 7) with a 10 adding point.

Comment 2: Outcomes and statistics

- What kinds of outcomes are expected? It's stated that "outcomes will be divided into two domains of the ICF-CY framework: (1) body function and structure and (2) activity and participation." but what are these likely to be measurements of specifically?
- How will different types of outcomes be analysed together?
- More detail needs to be given regarding the statistical analyses - particularly in relation to the stated limitation of the study "We anticipate that meta-analysis could be challenging due to the methodological heterogeneity of feasibility studies."

Response: In methodological part, we have indicated and given more details in outcomes and statistics.
- We provided a brief explanation and the example of outcome measurements in both (1) body function and structure and (2) activity and participation according to the ICF-CY framework, and the previous studies of the assessment of upper extremity function in children and adolescents with CP9 66-68 (p. 9-10, references added)
- The similar outcomes will be analysed regarding the specific domain and scale of outcome measurement. We have explained as “The risk ratio (RR) and odds ratio (OR) will be calculated for dichotomous outcome data. For continuous outcome data, the mean difference (MD) will be used to compare data measured on the same scales while the standard mean difference (SMD) will be used to summarize data measured on the different scales.” (p. 12-13)
- We have more clarified this statement in data synthesis and analysis as recommended. If there are sufficient data, we will use the inconsistency index (I2) statistic for heterogeneity test of similar outcomes and pool the effect sizes in a meta-analysis (p. 13).

Comment 3: Generally, the standard of written English is acceptable, but there are a few instances where sentences contain errors or have unclear meanings. A non-exhaustive list of examples:
- "Nonetheless, while the independence in activities and social participation of children and adolescents with CP is inevitably affected by motor dysfunction,10 11 eighty percent of those having motor impairments of upper extremity" (end of the sentence doesn't follow from the start)
- "Summarize of these reviews laid understanding..." (Summaries)
- "Nevertheless, the extent of service in the perspective of clinical intervention is still unclear." (service seems like the wrong word to use here, so the meaning of the sentence is unclear).

Response: We apologize for these errors. We have checked the sentences and corrected the unclear meanings as follows:
- We did a correction to “For upper extremity motor performance, there is evidence that up to eighty percent of CP clients have upper limb involvement and demonstrate limited functions.” (p. 5)
- We changed a word in this sentence to “Summaries” for correct grammar. (p. 7)
- We modified the text to “However, the extent of service in the perspective of clinical intervention still requires a broader scope to assemble knowledge on robotic-assisted therapy and provide explicit information on complementary and alternative approaches for upper extremity function in people with CP.” (p.7)

Reviewer 2: Dr. James Slaughter
Comment 1: The article clearly describes a protocol to be followed for a systematic review of robotic-assisted therapies for individuals with CP.

Response: Thank you very much for your comments.

Comment 2: If the authors have done an initial search, it would be useful to know approximately how many potential studies have been identified for inclusion from the searches. While this initial number

will likely be decreased after thorough examination of each study, it would be useful to know the scope of the initial search is neither too large or too small.

Response: We apologize for could not estimate the number of potential studies in our present systematic review due to the preliminary search is still in the process. However, based on the three previous literatures on robotic-assisted therapy for upper extremity function among children and adolescents with CP in the past five years, the number of final included articles have ranged from 9 to 35 studies.59-61

In 2016, Chen et al. have included 9 studies from a total of 159 papers.59 Equal to the review of Bayon et al. in the same year, the authors have found 9 included studies from a total of 226 articles.60 Latest research in 2019, Falzarano, et al. have identified 35 included studies from a total of 131 papers in their review.61

Reviewer 3: Dr. Hans Forssberg
Comment 1: The authors are planning a systematic review on robotic assisted therapy for upper extremities in children/youth with all types, and all severity levels of cerebral palsy. The study protocol is well designed and follow the book with all required information, and the manuscript is well written. I cannot find any flaw.

Response: Thank you very much for your comments.

Comment 2: My only concern is the introduction, which lacks clinical context. Unfortunately, this is quite common in medical engineering.

Response: We agree with your concern and have also modified the text in 2 statements.

- “Although the use of robots to improve functioning of upper extremities shows promise, the level of evidence in the pediatric group appears less than in adults.” (p. 6)

- “However, the extent of service in the perspective of clinical intervention still requires a broader scope to assemble knowledge on robotic-assisted therapy and provide explicit information on complementary and alternative approaches for upper extremity function in people with CP.” (p.7) This point is in accordance with the comments and suggestions provided by reviewer 1.

Comment 3: The introduction describes the CP condition from a rather “high level” with proper references, but gives little information about the clinical symptoms and functional limitations that the robot-assisted intervention is supposed to address.

Response: We consider the reviewer comment. We have compiled information about robotic-assisted therapy in several clinical groups and added more references of the effects of therapeutic robot in CP conditions as following statements:
- “Robotic devices have been widely integrated into clinicians’ service in order to mediate and assist interventions to enhance clients’ recovery and clinical outcomes resulting from stroke, traumatic brain injuries, multiple sclerosis, spinal cord injury, and other congenital and acquired movement disorders.”26-28 (p. 5)
- “From previous studies, the use of robotic devices has been found to improve kinematics,37-40 range of motion,41 42 muscle tone,43 44 postural control,45 46 and functionality of upper38 40 and lower extremities47 48 among individuals with CP.” (p. 5, reference added)

Comment 4: There is neither any information on other therapies used to improve motor function in the

upper extremities, although robots are often used in combinations with other modes of intervention.

Response: In response to your comment, we addressed other interventions for upper extremity function in children and adolescents with CP as referred.

- “To date, varying interventions are recommended to ameliorate movements and activities in upper extremity for CP clients, for example constraint-induced movement therapy (CIMT), hand-arm bimanual intensive therapy (HABIT), neurodevelopmental therapy (NDT), intramuscular injection of botulinum neurotoxin A (BoNT-A), as well as, robotic-assisted therapy.”36 53-56 (p. 6, reference added)

Comment 5: In particular, it would be important to mention that much of the evidence based interventions are based on motor learning and intensive, repetitive self-initiated movement, and in which way the use of robots can contribute to improvement.

Response: We added the statement emphasizing the theoretical basis of robotic-assisted therapy and application in order to improve clients’ functioning.

- “Based on the applying principles of neural plasticity, motor control, and motor learning, robotic rehabilitation premises in the adaptability of the brain, which may occur spontaneously over time and/or be induced by movement practice after lesion. Thus, the implementation of robotics primarily focuses on functional motor performance by providing intensive repetitive training, sensorimotor integration, and cognitive engagement through goal directed tasks to address the underlying symptoms and related problems due to client’s neurological conditions.”34-36 (p. 5, reference added)

VERSION 2 – REVIEW

REVIEWER	Jones, Christopher Brighton and Sussex Medical School, Medical Statistics
REVIEW RETURNED	15-Dec-2020

GENERAL COMMENTS	My previous comments have all be addressed appropriately.

REVIEWER	Forssberg, Hans Karolinska Institutet, Women's and Children's Health
REVIEW RETURNED	16-Dec-2020

GENERAL COMMENTS	The authors have done their best to answer my concerns regarding the introduction, which now has improved. I also think additional revisions have strengthened the quality and readability of the paper.